

ProACT: Proprioceptive Active Contact Sensing for Whole-Body Contact Localization on Humanoids

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Abstract—Proprioception-only estimation of external contact on humanoids has recently reached whole-body wrench prediction, but its localization accuracy is fundamentally limited: for contacts on the legs and torso during quiet stance the external force lies largely in the null space of the contact Jacobian and is therefore *unobservable* from a fixed posture. We observe that contact observability is a function of the robot’s motion, and introduce ProACT, a proprioceptive *active* contact-sensing framework: upon detecting a *sustained* contact whose location is ambiguous, the robot executes a small, deliberate probing motion chosen to make the previously-unobservable force component observable, and refines its estimate from the resulting proprioceptive sequence. ProACT couples an always-on passive base estimator with an on-demand active refinement stage, and is trained in simulation with ground-truth contact labels. **[TODO: one-sentence headline result: e.g. active probing raises leg/trunk localization from X% (passive) to Y%, matching/exceeding the passive ceiling.]**

Index Terms—humanoid robots, contact estimation, proprioception, active perception, force sensing, whole-body control

I. INTRODUCTION

[TODO: write introduction]

II. RELATED WORK

A. Proprioceptive external-force / contact estimation

Model-based momentum observers recover external joint torques from dynamics residuals [1]. Learning-based methods now predict a whole-body wrench field from proprioception alone [2], and self-supervised free-space torque models enable label-free calibration on real robots [3]. **[TODO: expand; position ProACT as the active complement to these passive methods]**

B. Active and tactile perception

[TODO:]

C. Whole-body compliance and interaction

Impedance control [4] underlies compliant physical interaction; GentleHumanoid [5] learns whole-body upper-body compliance from proprioception without force/tactile sensors, but regulates a manually-set stiffness rather than actively sensing where contact occurs. **[TODO: expand]**

TABLE I
CONTACT LOCALIZATION: PASSIVE VS. ACTIVE PROBING.

Method	Exact-link (%)		± 1 -link (%)	
	Legs/Trunk	Arms	Legs/Trunk	Arms
Passive (static)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
ProACT (scripted)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
ProACT (learned)	[TODO:]	[TODO:]	[TODO:]	[TODO:]

III. PROBLEM FORMULATION AND OBSERVABILITY

$$\tau_{\text{ext}} = \mathbf{J}^\top(\mathbf{q}) \mathbf{f}, \quad \mathbf{f} \in \text{null}(\mathbf{J}^\top(\mathbf{q})) \Rightarrow \tau_{\text{ext}} = \mathbf{0}. \quad (1)$$

[TODO: state the observability claim + why motion helps; optional Gramian.]

IV. METHOD

A. Passive base estimator

[TODO:]

B. Active probing

[TODO:]

C. Online refinement decoder

[TODO:]

D. Sim-to-real calibration

[TODO:]

V. EXPERIMENTS

[TODO:]

VI. RESULTS AND DISCUSSION

[TODO:]

VII. CONCLUSION

[TODO:]

REFERENCES

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